Do Now · retrieval

Lesson 052 · Binary Search Trees with Node Objects

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What does LinkedList<T>.AddAfter use to place a new node?

Q2

What does HEAD mean for a linked list?

Q3

What is an invariant of a BST?

Q4

A binary search tree has root 10, left child 6 and right child 18. Insert 15, then 12, using the usual smaller-left, larger-right rule. Determine the parent and side of the new node 12.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Discover — A binary search tree (BST) keeps smaller keys in the left subtree and larger keys in the right subtree under a consistent duplicate-key policy.

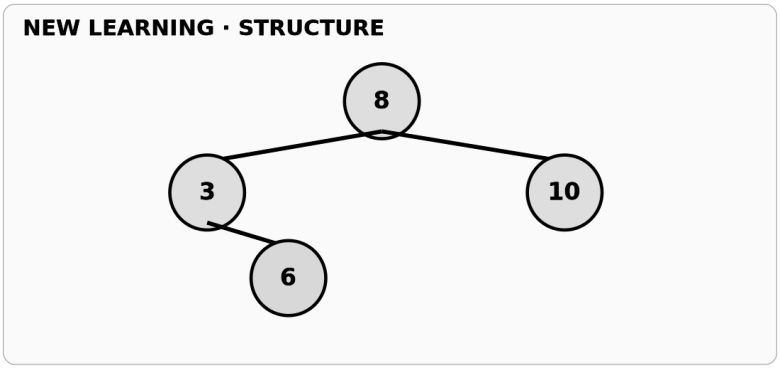
Core idea 2

Trace — Each node stores a key and optional Left/Right child references. Inserting begins at the root and follows comparisons until an empty child link is found.

Core idea 3

Evaluate — Inorder traversal (left, root, right) prints sorted keys for a valid BST without duplicates.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

BST — Binary search tree ordered by key.

root — Topmost tree node.

leaf — Node without child nodes.

inorder — Left subtree, root, right subtree.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

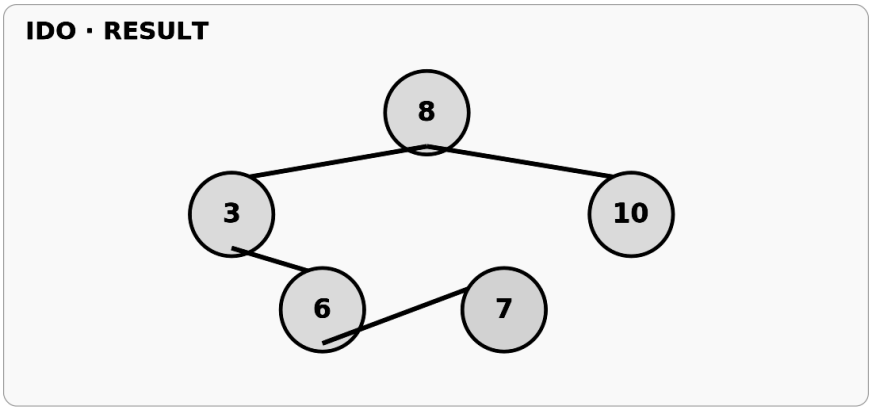
Problem

Insert key 7 into BST root 8, left 3, right 10, with 6 right of 3.

Follow the worked method

1. Explain the trace — 7 becomes right child of 6; follow comparisons until a null child is reached.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Explain the trace
Place 12 as left child of 14. Tree ordering remains valid.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

BST root8, left3, right10, insert 6. Show comparisons and location. [6]

Question 2

For BST with keys 8,3,10,6, give inorder traversal and explain sorting. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Explain why BST search may be O(log n) when balanced but O(n) when skewed. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

An unbalanced BST is guaranteed to have logarithmic search time.

Use key vocabulary from memory

1. Explain BST in your own words.

2. Explain root in your own words.

3. Explain leaf in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.